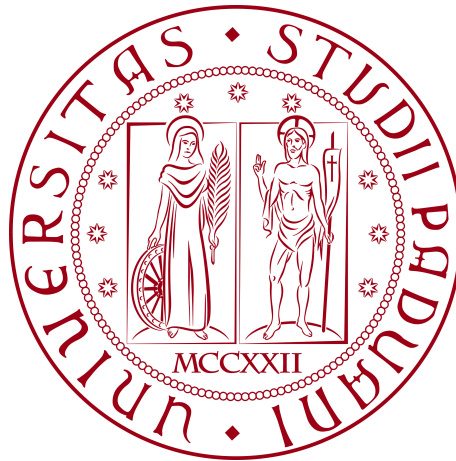


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When Does Structural Reasoning Help? Structural Generalization in Neurosymbolic Learning

Master's Thesis

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“We can only see a short distance ahead, but we can see plenty there that needs to be done.” — Alan Turing

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makes it difficult to predict where this field will lead in the future, but it also makes clear that there is still much to discover, understand, and build.

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Abstract

This thesis investigates the use of Structured Logic Tensor Networks (LTNs) for neuro-symbolic learning. The goal is to combine neural learning with explicit background logical knowledge. The proposed approach addresses an AI classification task with limited and heterogeneous training data. Entities and concepts are represented through learnable continuous embeddings. Logical relations and domain constraints are expressed using first-order logic. These logical formulas are converted into differentiable constraints using fuzzy logic semantics. The model jointly learns from labeled examples and the available background knowledge. A structured representation is used to organize concepts, relations, and logical dependencies. This allows domain knowledge to guide the learning process rather than relying only on observed data. The approach is evaluated against conventional neural network baselines. Experiments investigate both predictive accuracy and logical consistency of the learned model. Additional experiments analyze the effect of increasing or reducing the amount of background knowledge. The results examine whether logical constraints improve generalization, particularly with limited training data. The thesis also discusses computational and modeling challenges associated with differentiable logical reasoning. Overall, the study demonstrates the potential of Structured LTNs for integrating learning and reasoning in AI systems.

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Chapter 1

Introduction

Artificial intelligence systems are increasingly required not only to learn from data, but also to reason about the knowledge acquired from their environment. Modern machine learning methods have demonstrated remarkable capabilities in processing complex and high-dimensional data, including images, natural language, and other forms of unstructured information. However, learning directly from data does not by itself provide an explicit representation of knowledge or a principled mechanism for expressing logical constraints. Conversely, symbolic artificial intelligence provides powerful formalisms for representing and manipulating structured knowledge, but traditionally relies on explicitly defined symbols and rules and therefore faces difficulties when dealing with noisy, incomplete, or high-dimensional data. The integration of learning and reasoning has consequently become a central topic in neurosymbolic artificial intelligence [1]. Within this context, Logic Tensor Networks (LTNs) provide a framework for integrating first-order logic with differentiable neural computation. LTN introduces a many-valued, end-to-end differentiable logical language, called Real Logic, in which the elements of a first-order logic signature are grounded onto data through neural computational graphs. Logical formulas are assigned truth values in the interval $[0, 1]$, allowing logical constraints to participate directly in gradient-based learning [1]. This provides a common framework in which learning from data and reasoning with background knowledge can be expressed within the same formalism. More recently, the Structural Logic Tensor Networks (sLTN) framework extends the original LTN formulation to explicitly represent structural organization in data. While conventional LTN is primarily formulated for domains consisting of collections of individuals, many real-world datasets possess additional dimensions such as temporal order, sequence position, or graph structure. sLTN introduces

structural dimensions, structural variables, and structural relations as explicit components of the logical language, allowing such organization to be represented and reasoned about at the logical level [5]. This thesis investigates this extension from a research perspective, focusing not only on how structural information can be represented, but also on whether explicitly modelling such information can improve the generalization of neurosymbolic learning systems.

1.1 Logic Tensor Networks

Logic Tensor Networks were introduced as a neurosymbolic framework designed to combine the representational capabilities of first-order logic with the learning capabilities of neural networks. The central idea is to interpret the symbols of a logical language using differentiable tensorial representations, so that logical knowledge can be incorporated directly into a machine learning system [1]. Traditional machine learning approaches generally represent knowledge implicitly through numerical parameters learned from data. Such approaches are particularly effective when large amounts of labelled or unlabelled data are available, but the resulting representations do not necessarily expose the logical relationships that hold between concepts. Symbolic approaches, on the other hand, represent knowledge explicitly using concepts, relations, and logical rules. This makes symbolic representations suitable for reasoning and knowledge manipulation, but classical symbolic systems are generally less suited to directly processing raw, noisy, or high-dimensional observations [1]. Neurosymbolic approaches attempt to bridge this distinction by combining symbolic representations with sub-symbolic learning. In this setting, symbolic knowledge can provide constraints and relationships that guide learning, while neural models provide mechanisms for learning representations from data. A fundamental issue in such systems is therefore the grounding of symbolic entities into concrete data representations. LTN addresses this problem by assigning differentiable tensor representations and functions to the elements of a first-order logical language [1]. The LTN framework is based on Real Logic, a many-valued and differentiable extension of first-order logic. In this setting, logical formulas are not restricted to the classical truth values of true and false. Instead, their truth values are represented numerically

in the interval $[0, 1]$. Logical connectives and quantifiers are implemented using differentiable fuzzy-logic operations, allowing the satisfaction of logical formulas to be optimized using gradient-based methods [1]. This formulation leads to an important reinterpretation of learning. Rather than defining learning solely as the minimization of a task-specific numerical loss, LTN allows the learning problem to be expressed through a collection of logical formulas whose degree of satisfaction is optimized. The parameters of the underlying neural computational graphs can consequently be learned so as to maximize the overall truth value of the specified knowledge [1]. The resulting framework can express a variety of learning and reasoning tasks. The original LTN work demonstrates applications including data clustering, multi-label classification, relational learning, query answering, semi-supervised learning, regression, and embedding learning [1]. The common feature of these tasks is that data-driven learning and logical constraints can be expressed within a single differentiable framework. An LTN model can therefore be viewed as consisting of two interacting components. On the one hand, neural computation provides continuous representations and differentiable mappings from data to these representations. On the other hand, the logical layer specifies relationships and constraints between the represented entities. The grounding connects these two levels by mapping logical terms and predicates to differentiable tensor operations [1]. This combination is particularly relevant when background knowledge is available. Instead of treating the available knowledge as an external post-processing mechanism, LTN incorporates it directly into the learning objective. The logical structure can consequently influence the learned representations while the neural components provide the flexibility required to learn from numerical data [1].

1.2 From Flat Domains to Structured Data

Despite the generality of the LTN framework, its original formulation is primarily oriented towards domains that can be described as collections of individuals and relations between them. This setting is appropriate for many relational learning problems, where entities can be represented independently and logical predicates describe their properties or relationships. However, a large class of modern datasets contains an additional structural organization that

cannot be naturally described only through relations between independent individuals [5]. Examples include temporal sequences, where observations are associated with ordered time steps; sequential data, where the position of an element within a sequence is significant; and graphs, where the connectivity between nodes provides essential information about the domain. In these settings, the position or structural context of an observation is itself meaningful. Treating such information merely as an additional property of an individual does not necessarily provide a natural representation of the underlying structure [5]. Consider a temporal dataset represented as a tensor. One axis may correspond to different individuals or samples, while another axis represents time. The temporal axis has a meaning that is different from the identity of the individuals contained in the tensor. Similarly, in a sequence, the position of an element is not simply another individual in the domain: it represents its location within an ordered structure. In a graph, the organization of nodes and their connectivity introduces another form of structure that is not captured solely by treating nodes as an unordered collection [5]. This distinction motivates the introduction of explicit structural information into the logical language. The sLTN framework addresses this problem by extending LTN with structural dimensions that represent named dimensions of structured objects. Examples of structural dimensions include time steps, sequence positions, and graph nodes [5]. Structural dimensions are therefore conceptually different from ordinary logical sorts and individuals. An ordinary first-order variable ranges over individuals in a domain, whereas a structural variable ranges over positions associated with a structural dimension. This distinction allows logical formulas to refer explicitly to the organization of data rather than encoding structural information indirectly [5]. The introduction of structural dimensions also makes it possible to define structural relations. A structural relation can describe relationships between positions within a structure. For example, a relation such as

$$\text{next}(t, t')$$

can express that t' is the position immediately following t in a temporal or sequential dimension. Such relations can then be used inside logical formulas to express constraints involving

neighbouring or otherwise structurally related positions [5]. This extension is important because structural relationships can influence the interpretation of a formula. A constraint may not simply state that a property holds for an individual, but that it holds at a particular structural position, or that properties at two positions are related according to the structure of the underlying data. sLTN consequently provides a way to express temporal, sequential, and relational constraints directly within the logical language [5].

1.3 Structural Logic Tensor Networks

Structural Logic Tensor Networks extend the LTN framework by making structural dimensions first-class elements of the logical language. The framework introduces an explicit distinction between the logical domain of individuals and the structural organization associated with tensors. This allows logical expressions to refer to positions within structured data while retaining the differentiable fuzzy semantics of LTN [5]. The formal language of sLTN includes ordinary first-order elements such as sorts, constants, variables, functions, and predicates, together with structural elements such as structural variables and structural relations. Structural dimensions provide the link between the logical description and the axes of the underlying tensors [5]. A structural dimension can be understood as a named axis associated with a particular type of organization in the data. For example, a dimension called *time* may identify the temporal axis of a tensor, while another dimension may represent positions in a sequence. The structural dimension is not merely metadata: it participates in the interpretation of logical expressions by determining how tensorial data are aligned, selected, masked, or aggregated [5]. Structural variables are used to quantify over the positions of these dimensions. In contrast to ordinary first-order variables, which range over individuals, structural variables range over indices associated with structural dimensions. This distinction allows formulas to make explicit statements about the location of data within a structured object [5]. Structural relations provide a further level of expressiveness. They allow relationships between structural positions to be represented explicitly. For example, a temporal relation can identify consecutive time steps, while other structural relations can describe more general dependencies between positions. These relations can be Boolean or

fuzzy and can therefore be integrated into the differentiable semantics of the framework [5]. An important consequence of this design is that structural information becomes part of the logical theory rather than being handled exclusively by the underlying neural architecture. The logical language can therefore express constraints whose meaning depends directly on the organization of the data. This is particularly relevant for problems in which temporal or sequential dependencies are part of the knowledge that should guide learning [5]. The sLTN framework retains the central principle of LTN: logical expressions are interpreted through differentiable tensor operations. Thus, introducing structural dimensions does not replace the original tensorial semantics but extends it to account for additional axes of organization. The resulting framework can align logical expressions with tensor dimensions and use structural information when performing selection, masking, or aggregation operations [5]. An important theoretical property of sLTN is that the original LTN framework can be recovered as a special case. When no structural dimensions are introduced, the structural extension does not add additional structure to the interpretation, and the framework reduces to the original LTN setting [5].

1.4 Syntax, Semantics, and Tensorial Interpretation

A central aspect of sLTN is the explicit separation between the syntactic and semantic levels of the framework. The language first defines the symbols and expressions that are permitted by the theory. These expressions are then parsed and validated before being interpreted over tensorial data [5]. The signature specifies the vocabulary available to a theory. It contains the sorts, dimensions, constants, variables, functions, predicates, and structural relations required to construct formulas. This explicit signature provides a formal description of the entities and relationships that can appear in a logical theory [5]. The syntactic layer is followed by formula parsing and validation. This separation is important because it distinguishes the logical structure of an expression from the numerical representation used to evaluate it. A formula can therefore be treated as a symbolic object before being mapped to tensor operations by the interpretation mechanism [5]. At the semantic level, the symbols of the logical language are assigned tensorial interpretations. Constants and variables are

grounded into tensor representations, while predicates and functions are interpreted through differentiable operations. Logical connectives and quantifiers are then evaluated using fuzzy semantics, producing numerical truth values [1, 5]. Structural dimensions introduce an additional component into this interpretation. A structural dimension determines an axis of the tensorial representation, while structural variables identify positions along that axis. Structural relations determine relationships between such positions. The interpretation can consequently use the structural information to align tensors and to perform operations that respect the organization encoded by the data [5]. This tensorial interpretation is essential for maintaining differentiability. Logical reasoning is not implemented as a separate symbolic process disconnected from learning. Instead, formulas are translated into differentiable numerical operations whose outputs can contribute to an optimization objective. As in LTN, the truth value of a formula can therefore influence the learning of the parameters involved in its grounding [1, 5]. The separation between syntax and semantics also provides a modular view of the framework. The logical theory specifies what can be expressed, while the interpretation specifies how those expressions are connected to the underlying data. This distinction becomes particularly useful for structured data because the same logical concepts can be associated with different tensorial organizations through their structural dimensions [5].

1.5 Motivation and Research Gap

The introduction of structural dimensions and relations provides a mechanism for representing information that is intrinsic to the organization of structured data. However, the availability of such a representation does not by itself establish whether explicit structural reasoning provides a measurable advantage during learning. In particular, it remains important to determine the conditions under which structural information can improve the ability of a neurosymbolic model to generalize learned knowledge to structural configurations that were not observed during training. This question is particularly relevant because a model may achieve good predictive performance without explicitly representing structure. A purely neural model may learn statistical regularities from the observations, while a conventional

LTN may incorporate logical constraints without explicitly modelling structural positions and relations. Consequently, any improvement observed with sLTN should not automatically be attributed to the presence of explicit structural reasoning. This thesis therefore focuses on the relationship between structural information and generalization. Rather than assuming that explicit structural reasoning is beneficial in every setting, the study investigates whether its contribution depends on the extent to which the target task relies on structural relationships. In addition, the study examines whether this relationship changes when the amount of available training data is reduced.

The central research problem can therefore be formulated as follows:

When does explicit structural reasoning improve the generalization of neurosymbolic models to previously unseen structural configurations?

The experimental analysis is designed to identify conditions under which explicit structural representations provide a measurable benefit, as well as conditions in which their contribution is limited. This allows the study to characterize the role of structural reasoning rather than simply comparing the overall predictive performance of different models.

1.6 Research Questions

The main objective of this thesis is to investigate the conditions under which explicitly representing structural information can improve the ability of neurosymbolic models to generalize beyond the structural configurations observed during training. To address this objective, the thesis considers the following research questions:

- **RQ1 — Structural Generalization.**

To what extent does explicit structural reasoning affect the generalization of neurosymbolic models to previously unseen structural configurations?

- **RQ2 — LTN and sLTN Comparison.**

How does the generalization performance of sLTN compare with that of conventional LTN when the structural configurations encountered during testing differ from those observed during training?

- **RQ3 — Conditions for Structural Benefit.**

How does the benefit of explicit structural reasoning vary with the structural dependency of the task and the amount of available training data?

These research questions are addressed through a controlled experimental evaluation involving a neural baseline, a conventional Logic Tensor Network, and a Structural Logic Tensor Network incorporating explicit structural dimensions and relations.

1.7 Research Hypotheses

Based on the research questions introduced in the previous section, this thesis formulates the following hypotheses to be empirically investigated:

- **H1 — Structural Generalization.**

Explicit structural representations are expected to improve generalization to previously unseen structural configurations when the target task depends on structural relationships.

- **H2 — Structural Dependency.**

The relative benefit of explicit structural reasoning is expected to increase with the degree of structural dependency of the task. Conversely, when the target task can be solved primarily from local or non-structural information, the difference between models with and without explicit structural representations is expected to be smaller.

- **H3 — Data Efficiency.**

The benefit of explicit structural reasoning is expected to become more pronounced when the amount of available training data is reduced, particularly for tasks that strongly depend on structural relationships.

These hypotheses are formulated as empirical research hypotheses rather than assumptions about the superiority of sLTN. The experimental evaluation therefore considers both scenarios in which explicit structural reasoning provides a measurable benefit and scenarios in which its contribution is limited or absent. The objective is to identify the conditions associated with a structural advantage rather than to establish a priori that one model is universally superior to another.

1.8 Contributions of the Thesis

The primary contribution of this thesis is a systematic experimental investigation of the conditions under which explicit structural representations contribute to structural generalization in neurosymbolic learning. The experimental framework provides a controlled comparison between three model settings: a neural baseline, a conventional Logic Tensor Network, and a Structural Logic Tensor Network incorporating explicit structural dimensions and relations. This comparison provides a basis for separating the contribution of logical constraints from that of explicit structural representations. A further contribution is the design of controlled structural generalization scenarios in which the structural configurations encountered during evaluation differ from those observed during training. This experimental setting provides a direct means of assessing whether models can generalize learned knowledge across changes in structural organization. The thesis also investigates how the contribution of explicit structural reasoning changes with the structural dependency of the task. By considering tasks with different degrees of dependence on structural relationships, the study aims to characterize the conditions under which explicit structural representations become useful and the conditions under which their contribution is limited. Finally, the experimental evaluation considers different amounts of available training data. This analysis investigates whether the effect of explicit structural reasoning changes as supervision becomes more limited, with particular attention to tasks in which structural relationships are important for the target prediction.

1.9 Thesis Organization

The remainder of this thesis is organized as follows:

- **Chapter 2** introduces the theoretical background required for the study, including neurosymbolic learning, first-order logic, fuzzy logic, Logic Tensor Networks, and the principles underlying structured data representations.
- **Chapter 3** presents the Structural Logic Tensor Network framework in greater detail, focusing on its syntax, semantics, structural dimensions, structural variables, structural relations, and tensorial interpretation.
- **Chapter 4** describes the research methodology adopted in this thesis. It introduces the experimental task and dataset, the considered models, the structural configurations used during training and testing, the different levels of structural dependency, the amounts of training data considered, the evaluation metrics, and the experimental protocols used to address the research questions.
- **Chapter 5** presents the experimental evaluation of structural generalization. The chapter investigates the performance of the considered models on structural configurations observed during training as well as on previously unseen configurations. It also examines model performance under different levels of structural dependency and different amounts of training data.
- **Chapter 6** analyses the experimental results, with particular emphasis on the relationship between structural dependency, generalization performance, logical satisfaction, and the amount of training data required to achieve comparable performance.
- **Chapter 7** discusses the findings in relation to the research questions and hypotheses. Particular attention is given to identifying the conditions under which explicit structural reasoning provides measurable benefits, as well as the conditions under which its contribution is limited.

- **Chapter 8** summarizes the main findings of the thesis, discusses its limitations, and outlines directions for future research, including the potential extension of the experimental analysis to additional structured domains and neurosymbolic learning settings.

Chapter 2

Theoretical Background

This chapter introduces the theoretical concepts underlying the research presented in this thesis. It first discusses neurosymbolic artificial intelligence and the complementary roles of neural and symbolic approaches. The logical foundations required for Logic Tensor Networks are then introduced, followed by fuzzy and differentiable semantics. The chapter subsequently presents Logic Tensor Networks (LTN) and Structural Logic Tensor Networks (sLTN), with particular attention to the representation of structural information. Finally, the concepts of systematic generalization and structural dependency are introduced to motivate the research question investigated in this thesis.

2.1 Neurosymbolic Artificial Intelligence

Artificial intelligence has developed through different approaches to learning and knowledge representation. Sub-symbolic methods, particularly neural networks, learn numerical representations from data and have demonstrated strong capabilities in tasks involving complex and high-dimensional observations. Symbolic approaches, in contrast, represent knowledge explicitly through symbols, logical rules, and structured relationships. These approaches provide different advantages. Neural models can learn representations directly from examples, but their ability to generalize to configurations that differ systematically from those observed during training is not guaranteed [4]. Symbolic systems provide explicit representations of knowledge and relationships, but traditional symbolic methods are not naturally designed to learn directly from noisy or high-dimensional data.

Neurosymbolic artificial intelligence aims to combine these two paradigms by integrating

neural learning with symbolic knowledge and reasoning [garcez2022neurosymbolic]. The general objective is to exploit the representational and learning capabilities of neural models while retaining the ability to express structured knowledge explicitly. Logic Tensor Networks follow this approach by combining first-order logical representations with differentiable tensor computation. Logical symbols are grounded into numerical representations and differentiable functions, allowing logical knowledge to participate directly in learning [1].

This perspective is particularly relevant when the information required to solve a task includes relationships between observations. In such cases, the question is not only whether a model can learn from the observed data, but also whether it can represent and apply relationships to configurations that were not directly observed during training.

2.2 First-Order Logic

First-order logic (FOL) provides the formal language underlying Logic Tensor Networks. It allows knowledge to be expressed in terms of individuals, properties, relations, functions, and quantifiers. A first-order language contains non-logical symbols such as constants, functions, and predicates. Constants refer to individuals in a domain, while predicates express properties or relations involving these individuals. For example,

$$P(x)$$

can represent a property of an individual x , whereas

$$R(x, y)$$

can represent a relation between two individuals x and y . First-order logic also provides logical connectives and quantifiers.

The universal quantifier

$$\forall x$$

expresses that a formula holds for every relevant individual, while the existential quantifier

$$\exists x$$

expresses that it holds for at least one individual.

These elements can be combined into more complex formulas. For instance,

$$\forall x (P(x) \rightarrow Q(x))$$

expresses that every individual satisfying P also satisfies Q .

The relevance of first-order logic to neurosymbolic learning lies in its ability to represent relationships explicitly. Rather than considering observations independently, a logical theory can describe properties and relations that should hold across multiple individuals. LTN uses this logical structure while replacing classical truth evaluation with a differentiable many-valued interpretation [1].

2.3 Fuzzy Logic and Differentiable Semantics

Classical logic assigns binary truth values to formulas. LTN instead employs a many-valued semantics in which formulas receive values in the continuous interval

$$[0, 1].$$

A value close to 1 represents a high degree of truth, whereas a value close to 0 represents a low degree of truth. Intermediate values allow formulas to be partially satisfied. This continuous representation makes it possible to associate logical operators with differentiable functions. Logical connectives can be implemented through fuzzy operators, while quantified formulas can be evaluated through aggregation mechanisms over their grounded instances [1]. For a formula φ , its interpretation can therefore be written as

$$\mathcal{G}(\varphi) \in [0, 1].$$

Because the resulting operations are differentiable, the truth value of a logical formula can depend on learnable parameters and can consequently be incorporated into gradient-based optimization.

The specific choice of fuzzy operators is relevant to the behaviour of differentiable logical systems. Different operators can have different mathematical and optimization properties, which can affect the resulting learning dynamics [3].

In the context of this thesis, this observation motivates keeping the logical semantics fixed across the experimental conditions whenever possible. The objective is to investigate the effect of structural representation rather than differences caused by alternative fuzzy operators.

2.4 Logic Tensor Networks

Logic Tensor Networks (LTN) provide a framework for combining first-order logic with differentiable tensor computation. The framework, developed from earlier work on learning and reasoning with Logic Tensor Networks [6], provides differentiable interpretations of logical symbols and formulas [1].

The central mechanism is *grounding*. Logical symbols are associated with numerical representations or differentiable functions. A constant representing an individual can, for example, be grounded into a vector, while a predicate can be grounded into a function that evaluates the degree to which its arguments satisfy the corresponding property or relation.

For a unary predicate P , its grounding can be represented abstractly as

$$G(P) : \mathbb{R}^d \rightarrow [0, 1].$$

For a binary relation R , the corresponding grounding can be written as

$$G(R) : \mathbb{R}^d \times \mathbb{R}^d \rightarrow [0, 1].$$

The output represents the degree to which the corresponding predicate or relation is satisfied. Once logical symbols have been grounded, complete formulas can be evaluated using the differentiable semantics introduced above. Consequently, a logical theory

$$\mathcal{T} = \{\varphi_1, \dots, \varphi_n\}$$

can be associated with degrees of satisfaction for its constituent formulas. Learning can then be formulated as the optimization of an objective that encourages the desired logical constraints to be satisfied.

Conceptually, this can be expressed as

$$\max_{\theta} \text{Sat}_{\theta}(\mathcal{T}),$$

where θ denotes the learnable parameters and $\text{Sat}_{\theta}(\mathcal{T})$ represents the degree of satisfaction of the theory under the current interpretation.

This formulation allows logical knowledge to influence learning rather than being used only as a post-training verification mechanism. Neural representations and logical constraints are therefore optimized within the same differentiable framework [1].

2.5 Structural Information in Learning

Many learning problems involve data whose organization contributes to its meaning. A temporal sequence is a simple example. Given

$$x_1 \rightarrow x_2 \rightarrow x_3,$$

the observations are not merely an unordered collection: their positions define relationships between them. The same distinction appears in other structured domains, such as graphs,

where connections between entities form part of the information relevant to the task.

This creates a representational issue for learning systems. If the target concept depends on the organization of the observations, representing the individual observations alone may not provide an explicit representation of the relevant relationships.

The original LTN framework provides a mechanism for representing logical relations between individuals, but does not explicitly introduce structural dimensions such as sequence positions as separate elements of the logical language. Structural Logic Tensor Networks (sLTN) extend the framework in this direction by making structural dimensions explicit [5].

2.6 Structural Logic Tensor Networks

Structural Logic Tensor Networks (sLTN) extend LTN by treating structural dimensions as first-class elements of the framework. A structural dimension can represent an axis associated with the organization of data, such as temporal steps or sequence positions [5].

The framework introduces structural variables and structural relations in addition to ordinary logical variables and relations. While ordinary variables range over individuals, structural variables range over positions within a structural dimension. Structural relations can then describe relationships between these positions.

For a temporal structure, for example, a relation such as

$$\text{next}(t, t')$$

can represent the fact that position t' immediately follows position t [5].

This distinction allows logical formulas to refer explicitly to the organization of the data. A formula can therefore express not only that an entity has a certain property, but also that the property occurs at a position related to another position according to the structure being represented.

sLTN preserves the differentiable semantics of LTN while extending the representation with structural information. Importantly, when structural dimensions are not used, the framework can recover the original LTN setting [5]. This makes the two frameworks naturally

comparable in an experimental setting.

The introduction of structural representations, however, does not imply that they must improve performance in every task. If a target can be determined without considering structural relationships, explicitly representing those relationships may provide little additional information. The potential benefit of sLTN should therefore depend on the role that structure plays in the task.

2.7 Systematic Generalization

A central concept for the experimental setting of this thesis is *systematic generalization*. In supervised learning, a model can achieve high performance on test examples that resemble its training examples without necessarily demonstrating that it can apply learned relationships to novel configurations.

Lake and Baroni studied this issue using the SCAN benchmark, which evaluates sequence-to-sequence models on combinations of commands that can differ from those observed during training. Their results demonstrated that good performance on familiar configurations does not necessarily imply systematic generalization to novel combinations [4].

The distinction is relevant to the present work because structural generalization can be evaluated by deliberately separating the structural configurations encountered during training from those used at test time.

For example, a model could be trained using sequences such as

$$A \rightarrow B \rightarrow C$$

and

$$A \rightarrow B \rightarrow D,$$

while being evaluated on configurations such as

$$A \rightarrow C \rightarrow B \rightarrow D.$$

The individual elements may already be familiar to the model, while their structural arrangement is new.

In this thesis, *structural generalization* is used as an operational term for this experimental condition: the ability of a model to generalize from structural configurations observed during training to previously unseen configurations at test time.

This definition is intentionally specific to the experimental setting of the thesis. It provides a measurable distinction between performance on familiar structural configurations and performance on novel ones, without assuming that structural generalization constitutes a separate established theoretical category in the literature.

2.8 Structural Dependency

The potential usefulness of explicit structural representations depends on whether the target task actually requires structural information.

Consider two sequence-based classification tasks. In the first, the target depends only on which elements occur in the sequence. The order of those elements does not affect the correct output. In the second, the target depends on the relative position or ordering of elements. In this case, information about the sequence structure is necessary to determine the target. This thesis refers to this distinction as *structural dependency*. Structural dependency describes the extent to which solving a task requires information about relationships involving positions, order, or other structural properties of the input.

A task with low structural dependency can be solved primarily from properties of individual observations. A task with high structural dependency requires information about how those observations are organized.

This distinction is important for evaluating sLTN because it provides a controlled way of asking when explicit structural representation should matter. If structure is irrelevant to the target, an explicit structural representation is not expected to provide substantial additional information. Conversely, when the target directly depends on structural relationships, representing those relationships explicitly may provide a more direct correspondence between

the model representation and the task.

Structural dependency is therefore treated as an experimental factor rather than as an assumption about the superiority of one model over another.

2.9 Structural Generalization and Data Availability

The amount of training data provides a second factor that may affect the ability of a model to learn structural relationships. With more training examples, a model may encounter a larger variety of structural configurations. With fewer examples, it has less opportunity to learn such configurations directly from the data.

Logical constraints provide an alternative source of information because they can express relationships that are not limited to individual training examples [1]. However, the presence of such constraints does not automatically guarantee improved data efficiency. Their usefulness depends on their relevance to the task and on how they interact with the learning objective.

For this reason, training-data availability is treated as a secondary factor in the experiments. The primary focus remains structural generalization, while different amounts of training data are used to investigate whether the effect of explicit structural representation changes when fewer examples are available.

The relevant notion of data efficiency in this thesis is therefore statistical: the amount of training data required to obtain a given level of generalization performance. It does not refer to computational efficiency or training time.

2.10 Research Gap

The theoretical background presented above establishes three relevant observations.

First, Logic Tensor Networks provide a differentiable framework in which first-order logical knowledge can be combined with neural representations and used directly during learning [1].

Second, sLTN extends this framework by making structural dimensions and structural re-

lations explicit, providing a representation for information such as sequence positions and temporal relationships [5].

Third, research on systematic generalization shows that successful learning on familiar configurations does not necessarily imply successful generalization to novel configurations [4].

Taken together, these observations motivate a more specific empirical question. The availability of an explicit structural representation does not by itself establish when such a representation produces a measurable advantage in generalization.

In particular, it remains relevant to investigate whether the effect of structural representation depends on the relationship between the target task and the underlying structure, on the novelty of the structural configurations encountered at test time, and on the amount of training data available.

This thesis addresses this problem through a controlled comparison between a neural baseline, conventional LTN, and sLTN. The experiments vary structural dependency and the degree of structural novelty between training and testing, while training-data availability is considered as a secondary factor.

The objective is therefore not to assume that sLTN is universally superior to conventional LTN or to neural models. Instead, the objective is to determine the conditions under which explicit structural representation contributes to generalization and to characterize situations in which its contribution is limited.

Chapter 3

Structural Logic Tensor Networks

Structural Logic Tensor Networks (sLTN) extend the Logic Tensor Network (LTN) framework by introducing an explicit representation of the structural organization of data. While the original LTN framework provides a differentiable logical language for reasoning over collections of individuals, many learning problems involve additional organization such as temporal order, sequence position, spatial structure, or graph connectivity. In such settings, the relationships between positions in the data may be part of the knowledge that the model needs to reason about. sLTN addresses this aspect by making structural organization an explicit component of the logical language [5]. The central idea of sLTN is to introduce structural dimensions as first-class elements of the language. Structural dimensions correspond to named axes of structured data, such as time steps in a temporal sequence, positions in a sequence, or nodes in a graph. Structural variables can then range over these dimensions, while structural relations describe relationships between different structural positions. This allows logical formulas to refer not only to ordinary individuals, but also to the organization in which those individuals occur [5]. The framework preserves the main principles of LTN. Logical symbols are grounded into tensors or differentiable functions, while logical connectives and quantifiers are interpreted through differentiable fuzzy semantics. Consequently, logical constraints involving structural information can participate in gradient-based learning. In this sense, sLTN does not replace the LTN approach, but extends it to a richer class of structured representations [5].

3.1 Motivation and Limitations of Standard LTN for Structured Data

The original LTN framework is primarily designed for settings in which data can be represented as collections of individuals. Each individual belongs to a logical domain and can be associated with tensor representations, predicates, functions, and other logical symbols. This formulation is effective when the relationships between individuals are the main source of structure that needs to be represented. However, many datasets contain an additional organization that cannot be naturally described only in terms of independent individuals. Examples include temporal sequences, where observations are ordered in time; token sequences, where the position of an element is meaningful; and graphs, where the connectivity between nodes determines the structure of the data. In such cases, structural information is not simply an implementation detail of the tensor representation. It can influence the meaning of a logical statement itself [5]. For example, consider a sequence of observations indexed by time. A property holding at time t may be related to the property holding at the following time step t' . Expressing such a constraint requires the logical language to refer explicitly to the temporal relationship between the two positions. Similarly, in a graph, a property associated with one node may need to be related to the properties of its neighboring nodes. One possible approach is to encode these relationships indirectly using ordinary first-order individuals and predicates. However, this treats structural organization as part of the domain representation rather than as an explicit component of the logical language. sLTN instead introduces structural dimensions and structural relations so that these forms of organization can be directly represented and manipulated within logical formulas [5]. This distinction is particularly important for the study developed in this thesis. If structural organization can be explicitly represented in the logical language, it becomes possible to investigate whether such an explicit representation provides an advantage when a model is required to generalize to previously unseen structural configurations. The purpose of sLTN is therefore not only to represent structured data, but also to provide a formal mechanism through which structural information can participate in differentiable logical reasoning.

3.2 Structural Dimensions

A structural dimension is a named dimension associated with a particular organization of data. Typical examples include a temporal dimension representing time steps, a sequence dimension representing positions in an ordered sequence, or a graph dimension representing nodes. Structural dimensions therefore provide an explicit representation of the axes along which structured data are organized [5]. In sLTN, structural dimensions are part of the formal signature of the language. The signature extends the standard components of a many-sorted first-order language with a set of structural dimensions and structural variables. The dimensional profile associated with a symbol specifies which structural dimensions are relevant to its tensorial representation. This mechanism establishes a direct relationship between the logical representation and the underlying tensor representation. A tensor may contain one or more axes corresponding to structural dimensions, and these axes are explicitly named and tracked by the framework. Structural information can consequently be used when expressions are aligned, selected, masked, or aggregated. For example, suppose that a video is represented as a sequence of frames. A temporal dimension T can be associated with the sequence of frames. A tensor representing a batch of videos may then contain an axis corresponding to T . Rather than treating this axis as an anonymous tensor dimension, sLTN associates it with a named structural dimension that can be referenced by logical expressions. This representation is important because the same tensor shape can have different meanings depending on the organization represented by its axes. An axis representing time is conceptually different from an axis representing image width, even if both are represented numerically as tensor dimensions. sLTN therefore makes the relevant structural organization explicit within its language and semantics [5].

3.3 Structural Variables

Structural variables are used to refer to individual positions of a structural dimension. They are distinct from ordinary first-order variables. An ordinary first-order variable ranges over individuals belonging to a declared logical sort. For example, a variable x of sort *Image*

may range over images in a dataset. A structural variable, in contrast, ranges over positions associated with a structural dimension. If t is associated with a temporal dimension, it can represent a particular position in time. This distinction allows a formula to simultaneously refer to an individual and to the structural position in which that individual is observed. For example, a property can be associated with an object x at time t , and another property can be associated with the same object at a different time t' . Structural variables can also be quantified. Consequently, the language can express statements over all or selected positions of a structural dimension. This is particularly useful when the logical constraint depends on the organization of multiple positions rather than on a single observation. The distinction between first-order and structural variables is therefore one of the fundamental elements introduced by sLTN. First-order variables describe the entities being reasoned about, whereas structural variables describe where those entities occur within an explicitly represented structure [5].

3.4 Structural Relations

Structural dimensions provide a way to identify positions, but structured reasoning also requires a mechanism for expressing relationships between those positions. sLTN introduces structural relations for this purpose. A structural relation connects elements of one or more structural dimensions. These relations may represent properties such as temporal adjacency, sequence ordering, or graph connectivity. They are interpreted as Boolean or fuzzy masks over structural dimensions and can therefore be integrated into the differentiable semantics of the framework [5]. A simple example is the temporal relation $next(t, t')$, which identifies pairs of consecutive positions in a temporal dimension. This relation can be used to express a constraint such as

$$\forall t, t' \mid next(t, t') : (A(x, t) \rightarrow A(x, t')) .$$

The formula states that whenever the relation $next(t, t')$ holds, the property A should be preserved from one temporal position to the next. The important aspect of this formulation

is that temporal adjacency is represented directly by the structural relation rather than being implicitly encoded by the neural architecture [5]. Structural relations are therefore the mechanism through which dependencies between positions become accessible to logical reasoning. The same principle can be applied to other structures. For a graph, for example, a relation could identify neighboring nodes. For a sequence, it could represent consecutive positions or another predefined ordering. Because structural relations can be Boolean or fuzzy, they can also be incorporated into the differentiable semantics of sLTN. This allows structural knowledge to be used together with learned representations rather than requiring a completely symbolic preprocessing stage.

3.5 Syntax and Semantics

The formal language of sLTN extends a many-sorted first-order signature with structural components. The signature contains sorts, dimensions, constants, individual variables, functions, predicates, structural variables, and structural relations. In addition, typing information specifies the structural dimensions associated with symbols and the dimensions consumed or produced by functions and predicates [5]. The framework separates the syntactic description of a theory from its tensorial interpretation. Symbols are first declared in a signature. Formulas are then constructed and checked against this signature before concrete tensor representations are assigned through an interpretation. This separation is relevant for two reasons. First, it makes the logical structure of a theory explicit independently of the particular tensors used during learning. Second, it allows structural information to be checked at the level of the language. Function and predicate applications must be compatible with their declared sorts and structural profiles, while structural annotations and quantifiers must refer to appropriate structural variables and dimensions [5]. The concrete syntax of sLTN provides logical operators and quantifiers in both symbolic and textual forms. In addition to ordinary first-order quantification, the language supports structural quantification. This allows formulas to quantify explicitly over positions belonging to structural dimensions. The resulting framework can therefore express formulas in which ordinary logical entities and structural positions coexist. A formula can describe a property of an individual, constrain

the property across different structural positions, or condition a statement on a structural relation between those positions.

3.6 Tensor Representation and Named Axes

A central characteristic of sLTN is that structural information is directly connected to tensor dimensions. The framework interprets terms and formulas as annotated tensors whose axes may correspond to ordinary variables, structural dimensions, and domain dimensions [5]. Consider again a collection of videos represented by a tensor. If a batch contains 32 videos, each consisting of 8 frames, the corresponding tensor may contain a batch axis, a temporal structural axis, and image-domain axes. The temporal axis is explicitly associated with a structural dimension such as T . The use of named structural axes allows expressions to be aligned according to their semantic dimensions rather than relying only on their numerical tensor positions. This is particularly relevant when formulas combine expressions carrying different structural profiles. Structural operations can therefore include alignment, broadcasting, selection, masking, and aggregation along explicitly identified axes. The resulting semantics provides a bridge between the logical representation of structure and its tensorial implementation. This design also preserves compatibility with ordinary LTN representations. If no structural dimensions are associated with an expression, the corresponding tensorial interpretation reduces to the unstructured case. More generally, sLTN is designed so that when structural dimensions are absent, the original LTN semantics is recovered as a special case [5].

3.7 Logical Reasoning over Structural Information

The introduction of structural dimensions and structural relations extends the types of constraints that can be expressed within the logical language. In standard LTN, logical formulas can describe relationships between individuals and constrain their learned representations. sLTN additionally allows these formulas to refer to the organization of the underlying data. This makes it possible to express constraints involving temporal persistence, sequence de-

dependencies, or relationships between connected structural positions. A structural relation acts as a logical condition identifying which combinations of positions should be considered related. For example, let $A(x, t)$ denote a property of an individual x at temporal position t . A persistence constraint can be expressed using a relation $next(t, t')$:

$$\forall t, t' \mid next(t, t') : (A(x, t) \rightarrow A(x, t')) .$$

The structural relation determines where the logical implication is applied. Consequently, the formula does not simply require A to hold independently at all positions. Instead, it specifies a relationship between the truth of A at two structurally connected positions. The same mechanism can represent more general structural dependencies. For instance, if a property depends on the order of elements in a sequence, the logical theory can refer to the corresponding structural positions and relations between them. In a graph setting, a relation describing adjacency can determine which nodes participate in a particular logical constraint. The resulting representation provides a direct way to combine learned representations with explicit structural knowledge. Logical constraints can still be optimized through differentiable fuzzy semantics, but the constraints themselves can now depend on the organization of the structured input [5].

3.8 Temporal Structures as a Case Study

Temporal data provide a natural example of the motivation behind sLTN because temporal order is an intrinsic part of the representation. The sLTN paper uses a video classification example in which a temporal dimension represents the sequence of frames in a video [5]. In this setting, a video can be associated with a structural dimension T , whose positions correspond to individual frames. A frame can therefore be represented not only by its visual content but also by its position along the temporal dimension. A structural relation such as $next(t, t')$ can then identify consecutive frames. Logical formulas can use this relation to impose constraints across adjacent positions. For example, if a property is expected to persist through consecutive frames, this dependency can be expressed explicitly using the temporal

relation. This example illustrates the main conceptual difference introduced by sLTN. The temporal organization is not left exclusively to the neural architecture. Instead, it becomes accessible to the logical language and can therefore be used when defining differentiable logical constraints. The temporal case is particularly relevant to the research question of this thesis because sequences provide a controlled environment in which structural organization can be manipulated independently from the underlying individual elements. This makes it possible to study whether explicitly representing structural information affects the ability of a neurosymbolic model to generalize to structural configurations that were not observed during training.

3.9 Relationship between LTN and sLTN

sLTN should be understood as an extension of LTN rather than as a separate alternative framework. Both approaches rely on the same general neurosymbolic principle: logical expressions are grounded into differentiable tensorial representations, and logical constraints are evaluated using fuzzy semantics so that they can participate in gradient-based optimization [1, 5]. The main difference concerns the representation of structure. Standard LTN provides a logical language for reasoning over individuals and their relationships, whereas sLTN introduces structural dimensions, structural variables, and structural relations as explicit components of the language. This distinction can also be understood in terms of the tensor representation. In standard LTN, tensor dimensions primarily represent the data required to ground logical entities. In sLTN, selected dimensions can additionally be given an explicit structural meaning. Their role can then be referenced by the logical language. An important consequence is that the original LTN setting remains available as a special case. When no structural dimensions are used, the additional structural machinery is absent and the framework recovers the corresponding unstructured LTN semantics [5]. This relationship is relevant for experimental comparison because it makes it possible to investigate the effect of explicit structural representation while keeping the underlying neurosymbolic paradigm consistent. The key question is therefore not whether sLTN constitutes a fundamentally different learning framework, but whether making structural information explicit produces

measurable differences when the learning problem depends on such information.

3.10 Limitations and Open Issues of sLTN

Although sLTN introduces a formal mechanism for representing structural information, the framework is presented as an initial framework and software library rather than as a fully validated solution for structured neurosymbolic learning. The authors explicitly identify the need for further stabilization, testing, and systematic empirical validation beyond the illustrative examples presented in the paper [5]. Several directions are identified as future work. One concerns the development of dedicated structured theories for concrete application domains, in which domain-specific objects, predicates, dimensions, and structural relations can be defined systematically. Another concerns extensions of the language with finitely recursive computational primitives while preserving differentiability and a well-defined tensor semantics. The paper also identifies the need for principled strategies for selecting or scheduling the power-mean exponent used in aggregation, with the objective of improving training stability and supporting more systematic hyperparameter selection. Further work is proposed on benchmarking the library components and comparing sLTN with alternative approaches for relational, temporal, spatial, and graph-based learning [5]. These open issues are particularly relevant to the present thesis. The introduction of explicit structural representations provides a formal mechanism for reasoning about structured data, but the empirical conditions under which this representation is actually beneficial remain an open question. In particular, the existence of a structural language does not by itself establish that explicit structural reasoning improves generalization or reduces the amount of data required for learning. The following chapter therefore moves from the formal framework to an empirical investigation. Rather than assuming that structural representations are always advantageous, the experimental methodology is designed to examine whether their effect depends on the degree to which the target task relies on structural information and on the availability of training data.

Chapter 4

Experimental Methodology

Actual research design: task, data, models, train/test structures, structural dependency, data amounts, metrics

Chapter 5

Experimental Results

What happened in the experiments — tables, plots, accuracy/F1, generalization results

Chapter 6

Analysis of Structural Generalization

Interpret the results specifically in terms of when structural reasoning helps

Chapter 7

Discussion

Answer RQs/Hypotheses, compare findings with LTN/sLTN literature, explain implications and limitations

Chapter 8

Conclusions and future research

Final findings, contribution, limitations, future research

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